Arijit Guha

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Education

Indian Institute of Technology Kharagpur 2026–2028 (expected)
M.Tech in Artificial Intelligence, First Year CGPA: Not yet published

Indian Institute of Technology (ISM) Dhanbad 2020–2024
B.Tech in Electronics and Communication Engineering CGPA: 8.03/10

Research Interests

Computer Vision and Deep Learning; Machine Learning;
Natural Language Processing and Large Language Models

Research and Technical Projects

Adaptive MambaVision – Input-Adaptive Feature Fusion GitHub
Under Prof. Mahesh Mohan M. R.

- Extended **MambaVision (CVPR 2025)** with token-dependent gates that adaptively combine state-space and convolutional features across six MambaVision-T mixers.
- Added only **5,132 parameters (∼0.016%)** and used identity initialization to preserve the pretrained model's outputs at initialization.
- Evaluated on CIFAR-100 and Oxford-IIIT Pet; observed **82.84% vs. 81.88% accuracy (+0.96 percentage points)** on an additional 5,000-image CIFAR-100 evaluation sample, while documenting mixed gains and computational overhead.

CleanQuad – Autonomous Quadruped Litter Collection GitHub
Under Prof. Prabhat Kumar Mishra

- Developed simulated Unitree A1 locomotion with trot gait generation, 3-D inverse kinematics, and yaw stabilization; passed **20/20 tests without falls**, with reported speed errors below **2%**.
- Generated **4,000 auto-labeled images** and fine-tuned YOLOv8n for litter detection, achieving **0.965 mAP@0.5** versus **0.342** for a color-based baseline on 500 unseen-layout test images.
- Integrated locomotion and perception components into a team-built simulation that collected **7/7 reachable litter items**.

One Pixel Attack – Black-Box Adversarial Evaluation GitHub
Under Prof. Koustav Rudra

- Implemented the **One Pixel Attack** using PyTorch and a custom NumPy differential evolution optimizer, without access to model gradients.
- Obtained untargeted attack success rates of **36.46% on ResNet-18** and **47.33% on VGG-16 BN** over 96 and 469 initially correctly classified CIFAR-10 images, respectively.
- Built evaluation tools for targeted and untargeted attacks, query counts, perturbation visualizations, and automated checks of the one-pixel constraint.

Research and Industry Experience

TEXMiN | AI/ML Intern IIT (ISM) Dhanbad | May 2023-May 2024

- Worked with the R&D team on hyperspectral image classification of **76 minerals** using **256-channel** spectral imagery for mineral deposit identification.
- Designed a workflow combining data augmentation, PCA/UMAP visualization, and **2D/3D CNN and Vision Transformer** models.

Brane Enterprise | Software Engineer Hyderabad | Aug 2024–Sep 2025

- Contributed to Enterprise Control Tower (ECT), developing Python/Django backend components for the **Stringer chatbot**.

Technical Skills

Programming: Python (high proficiency), C++(intermediate proficiency)
ML Frameworks & Libraries: PyTorch, NumPy, Pandas,Scikit-Learn,Matplotlib (intermediate proficiency)
Other Tools: MuJoCo, Django, Flask, Git

Relevant Coursework

- Foundations of Machine Learning
- Interpretable Machine Learning
- Mathematics and Algorithms for AI
- AI/ML for Robot Autonomy
- Computer Vision
- Natural Language Processing

Achievements

- **GATE:** All India Rank 269 out of 200K candidates
- **JEE Advanced:** All India Rank 4908 out of 1.3M candidates
- **Gold Medal**, Inter-IIT Tech Meet 11.0 – Saptang Labs Challenge.

Leadership

Division Head, CyberLabs Information Security Division.
Core Member, HackFest'22 Technical Team.